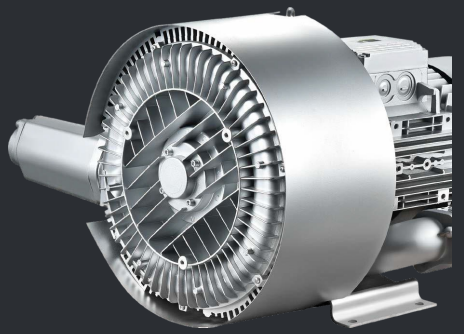
GB SERIES

Vortex air pump / blower pump / ring blower



POWER SUPPLY

3~ 380–415 V (models from 0.55 kW)

1~ 220–240 V (models up to 2.2 kW)

50 Hz

MODEL CODE

GB-2200S/2

S	three-phase 3~ (without S: single-phase 1~)
/2	double stage

VORTEX AIR PUMP

Single-stage and double-stage | 50 Hz

Redpump
Italy Standard

SINGLE STAGE

Power	**0.25 – 18 kW**
Max. flow	**up to 1050 m³/h**
Max. pressure	**up to 79 kPa**
Max. vacuum	**up to 40 kPa**
Speed	**n ≈ 2800 /min**

DOUBLE STAGE

Power	**0.55 – 15 kW**
Max. flow	**up to 560 m³/h**
Max. pressure	**up to 72 kPa**
Max. vacuum	**up to 57 kPa**

APPLICATIONS

BLOWING

Combustion air supply	Compost and wastewater aeration	Powder conveying
Bottle washing and drying	Pneumatic capsule transport	Incinerator air supply
Trim and scrap collection	Electroplating bath agitation	Exhaust gas dispersion
Air-knife drying	Aquaculture aeration	Vehicle drying
Air cushion handling	Molded-part release	Printing ink drying

SUCTION

Screen-printing vacuum hold-down	Printing press suction	Milk-carton filling
Welding fume and dust extraction	Industrial vacuum cleaning	Soybean conveying
Textile suction	Vacuum dewatering	Vacuum workholding
Film hold-down for photoengraving	Book trimming hold-down	Tunnel ventilation
Pneumatic conveying	Fabric cutting suction	Garment pressing steam extraction

OPERATING PRINCIPLE

The impellers are mounted directly on the motor shaft for compression without contact or friction. Maximum operational reliability, even at high differential pressure, is ensured by bearings located outside the compression chamber.



CONSTRUCTION

Vortex air pump exploded view and operating principle

PARTS

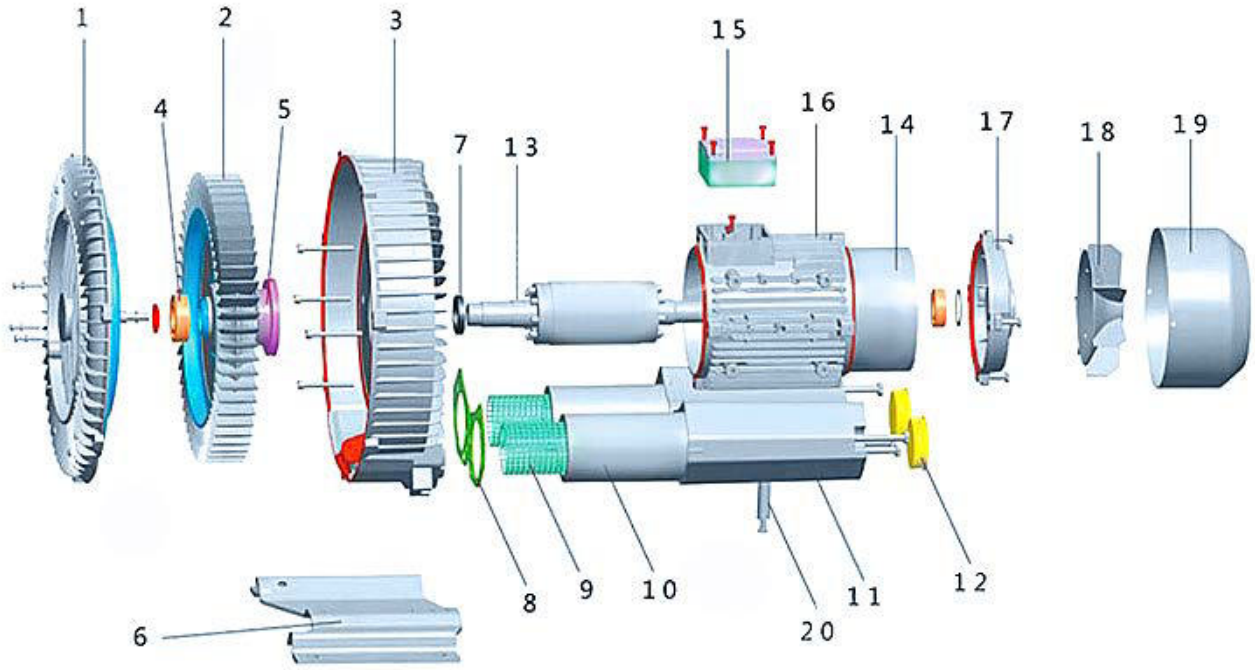
1	Pump cover
2	Impeller
3	Pump body
4	Bearing
5	Bearing seat
6	Base plate
7	Oil seal
8	Gasket
9	Silencer mesh
10	Silencer foam
11	Silencer pipe
12	Connector (flange)
13	Rotor
14	Motor
15	Terminal box
16	Motor housing
17	Motor end cover
18	Motor fan
19	Motor back cover
20	Support tube

CONSTRUCTION & OPERATING PRINCIPLE

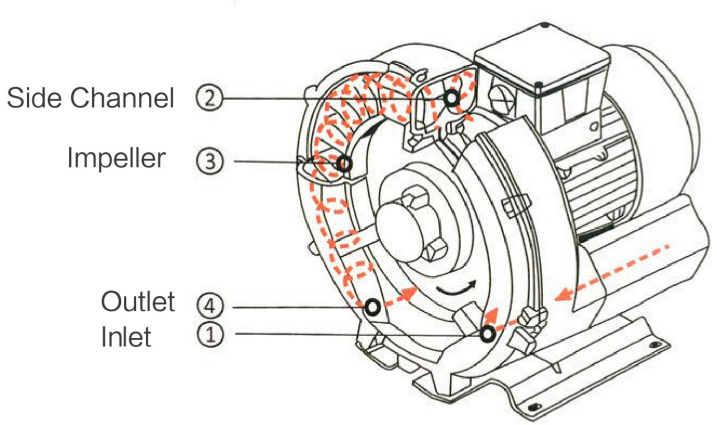
Exploded view

Redpump
Italy Standard

EXPLODED VIEW



OPERATING PRINCIPLE



The impellers are mounted directly on the motor shaft for compression without contact or friction. Maximum operational reliability, even at high differential pressure, is ensured by bearings located outside the compression chamber.

The gas is taken in through inlet 1. As it enters side channel 2, the rotating impeller 3 accelerates the gas in the direction of rotation. Every rotation adds kinetic energy, while centrifugal force moves the gas outward through the impeller blades and raises its pressure.

The pressure rises further along the side channel. The side channel narrows at the rotor, sweeping the gas off the impeller blades and discharging it through outlet silencer 4, where it exits the side channel blower.

GB SERIES

GB-250
GB-370

High-pressure vortex air pump



RANGE CHARACTERISTICS

Power	**0.25 – 0.37 kW**
Max. flow	**up to 60 m³/h**
Max. pressure	**up to 16 kPa**
Max. vacuum	**up to 11 kPa**
Speed	**n ≈ 2800 /min**
Noise	**< 56 dB**

MODEL CODE

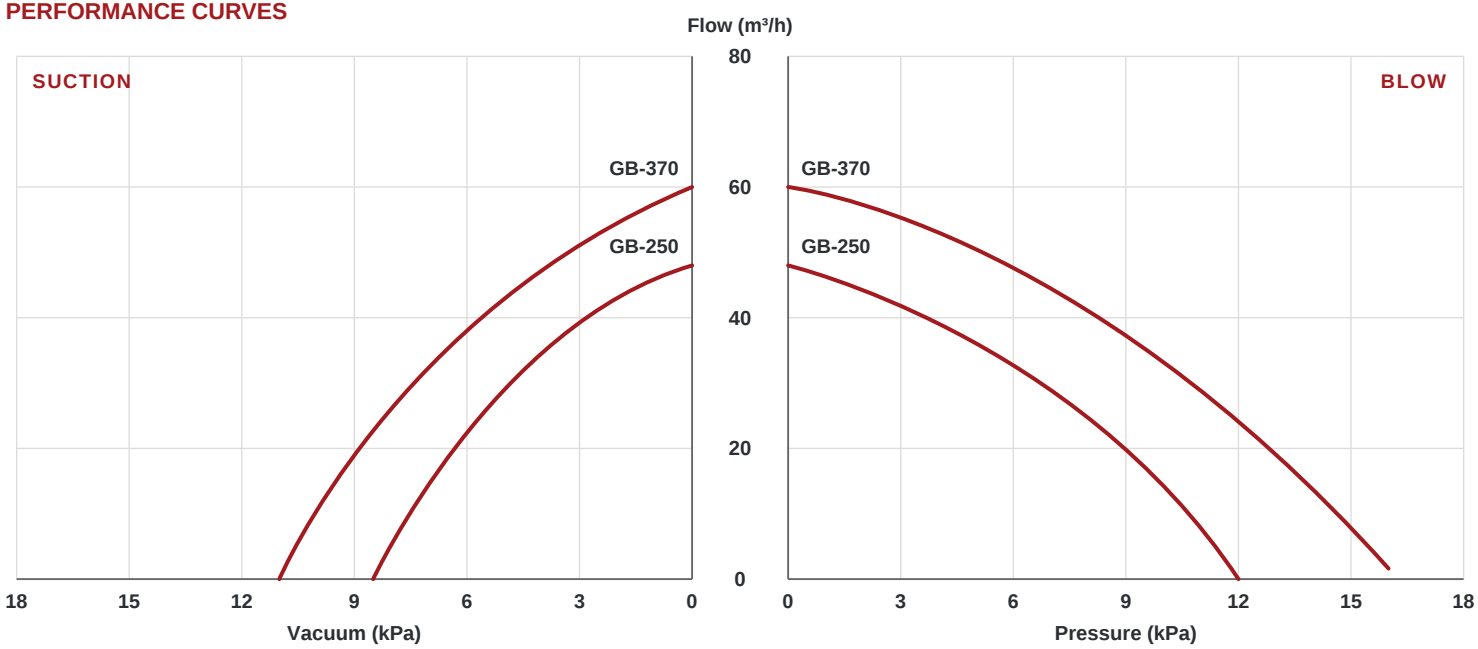
GB-250 · GB-370
Single-phase 1~ 220–240 V

PERFORMANCE & TECHNICAL DATA

50 Hz | n ≈ 2800 /min

Redpump
Italy Standard

PERFORMANCE CURVES



TECHNICAL DATA

MODEL	POWER kW	POWER HP	RATED CURRENT A	MAX. FLOW m³/h	MAX. PRESSURE kPa	MAX. VACUUM kPa	NOISE dB
SINGLE-PHASE 1~ 220–240 V							
GB-250	0.25	0.33	2.5	48	12	8.5	< 56
GB-370	0.37	0.5	3.5	60	16	11	< 56

OPERATING PRINCIPLE

The impellers are mounted directly on the motor shaft for compression without contact or friction. Maximum operational reliability, even at high differential pressure, is ensured by bearings located outside the compression chamber.

NOTES

Performance curves of the GB Series vortex air pumps are for reference only and are not a basis for testing.

Rated current at 220 V (1~).

GB SERIES

GB-550
GB-750

High-pressure vortex air pump



RANGE CHARACTERISTICS

Power	**0.55 – 0.75 kW**
Max. flow	**up to 145 m³/h**
Max. pressure	**up to 22 kPa**
Max. vacuum	**up to 18 kPa**
Speed	**n ≈ 2800 /min**
Noise	**< 60 dB**

MODEL CODE

GB-550S · GB-750S
Three-phase 3~ 380–415 V

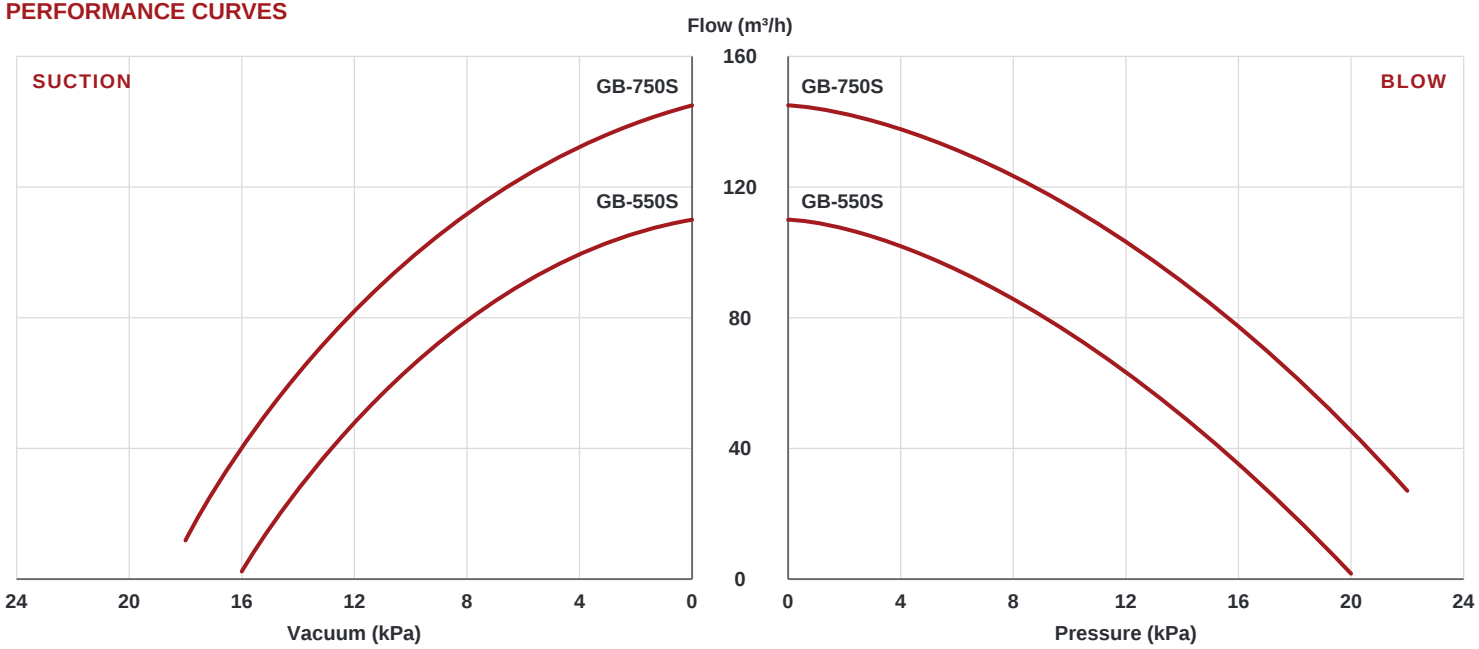
GB-550 · GB-750
Single-phase 1~ 220–240 V

PERFORMANCE & TECHNICAL DATA

50 Hz | n ≈ 2800 /min

Redpump
Italy Standard

PERFORMANCE CURVES



TECHNICAL DATA

MODEL	POWER kW	POWER HP	RATED CURRENT A	MAX. FLOW m³/h	MAX. PRESSURE kPa	MAX. VACUUM kPa	NOISE dB
THREE-PHASE 3~ 380–415 V							
GB-550S	0.55	0.75	1.45	110	20	16	< 60
GB-750S	0.75	1	1.81	145	22	18	< 60
SINGLE-PHASE 1~ 220–240 V							
GB-550	0.55	0.75	4.8	110	20	16	< 60
GB-750	0.75	1	6.3	145	22	18	< 60

OPERATING PRINCIPLE

The impellers are mounted directly on the motor shaft for compression without contact or friction. Maximum operational reliability, even at high differential pressure, is ensured by bearings located outside the compression chamber.

NOTES

Performance curves of the GB Series vortex air pumps are for reference only and are not a basis for testing.

Curves GB-550S / GB-750S also apply to GB-550 / GB-750 (1~).

Rated current at 220 V (1~) and 380 V (3~).

GB SERIES

GB-1100
GB-1500

High-pressure vortex air pump



RANGE CHARACTERISTICS

Power	**1.1 – 1.5 kW**
Max. flow	**up to 220 m³/h**
Max. pressure	**up to 26 kPa**
Max. vacuum	**up to 21 kPa**
Speed	**n ≈ 2800 /min**
Noise	**< 60 dB**

MODEL CODE

GB-1100S · GB-1500S
Three-phase 3~ 380–415 V

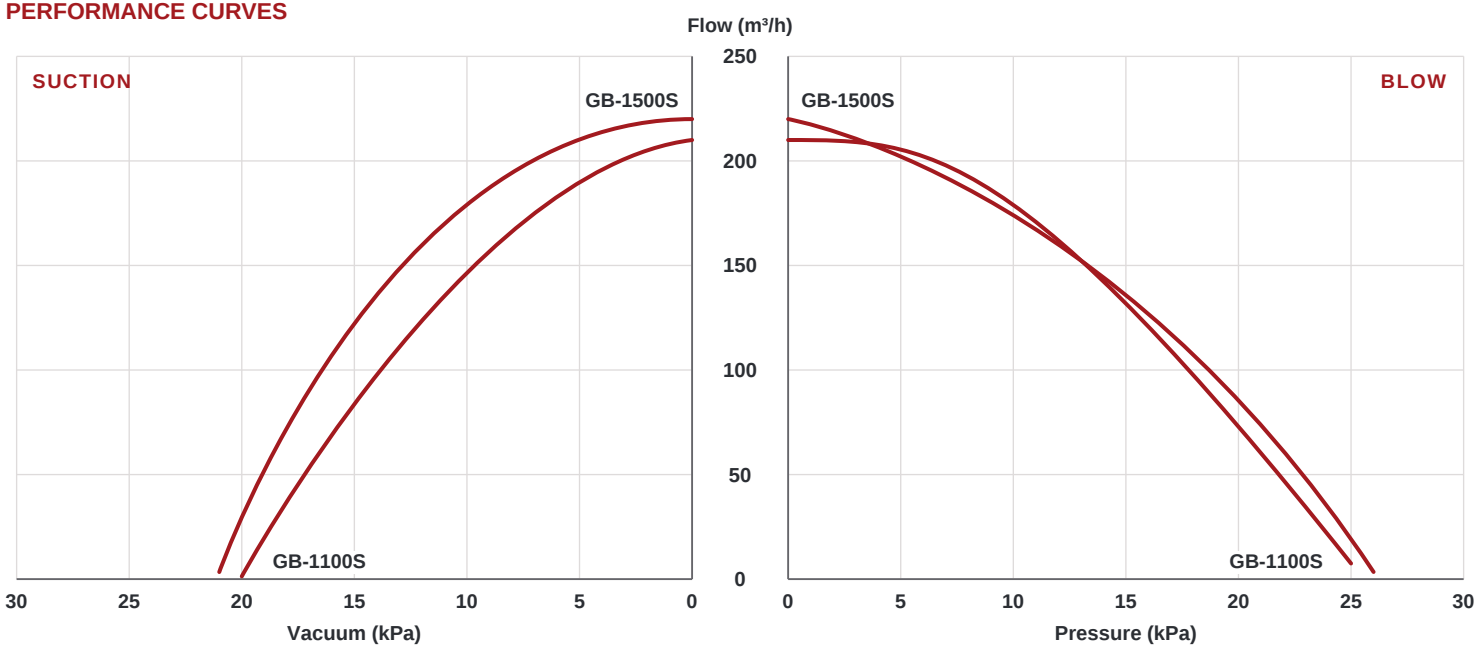
GB-1100 · GB-1500
Single-phase 1~ 220–240 V

PERFORMANCE & TECHNICAL DATA

50 Hz | n ≈ 2800 /min

Redpump
Italy Standard

PERFORMANCE CURVES



TECHNICAL DATA

MODEL	POWER kW	POWER HP	RATED CURRENT A	MAX. FLOW m³/h	MAX. PRESSURE kPa	MAX. VACUUM kPa	NOISE dB
THREE-PHASE 3~ 380–415 V							
GB-1100S	1.1	1.5	2.52	210	25	20	< 60
GB-1500S	1.5	2	3.44	220	26	21	< 60
SINGLE-PHASE 1~ 220–240 V							
GB-1100	1.1	1.5	8.7	210	25	20	< 60
GB-1500	1.5	2	11.8	220	26	21	< 60

OPERATING PRINCIPLE

The impellers are mounted directly on the motor shaft for compression without contact or friction. Maximum operational reliability, even at high differential pressure, is ensured by bearings located outside the compression chamber.

NOTES

Performance curves of the GB Series vortex air pumps are for reference only and are not a basis for testing.

Curves GB-1100S / GB-1500S also apply to GB-1100 / GB-1500 (1~).

Rated current at 220 V (1~) and 380 V (3~).

GB-2200

High-pressure vortex air pump



RANGE CHARACTERISTICS

Power	**2.2 kW**
Max. flow	**up to 325 m³/h**
Max. pressure	**up to 36 kPa**
Max. vacuum	**up to 28 kPa**
Speed	**n ≈ 2800 /min**
Noise	**< 70 dB**

MODEL CODE

GB-2200S
Three-phase 3~ 380–415 V

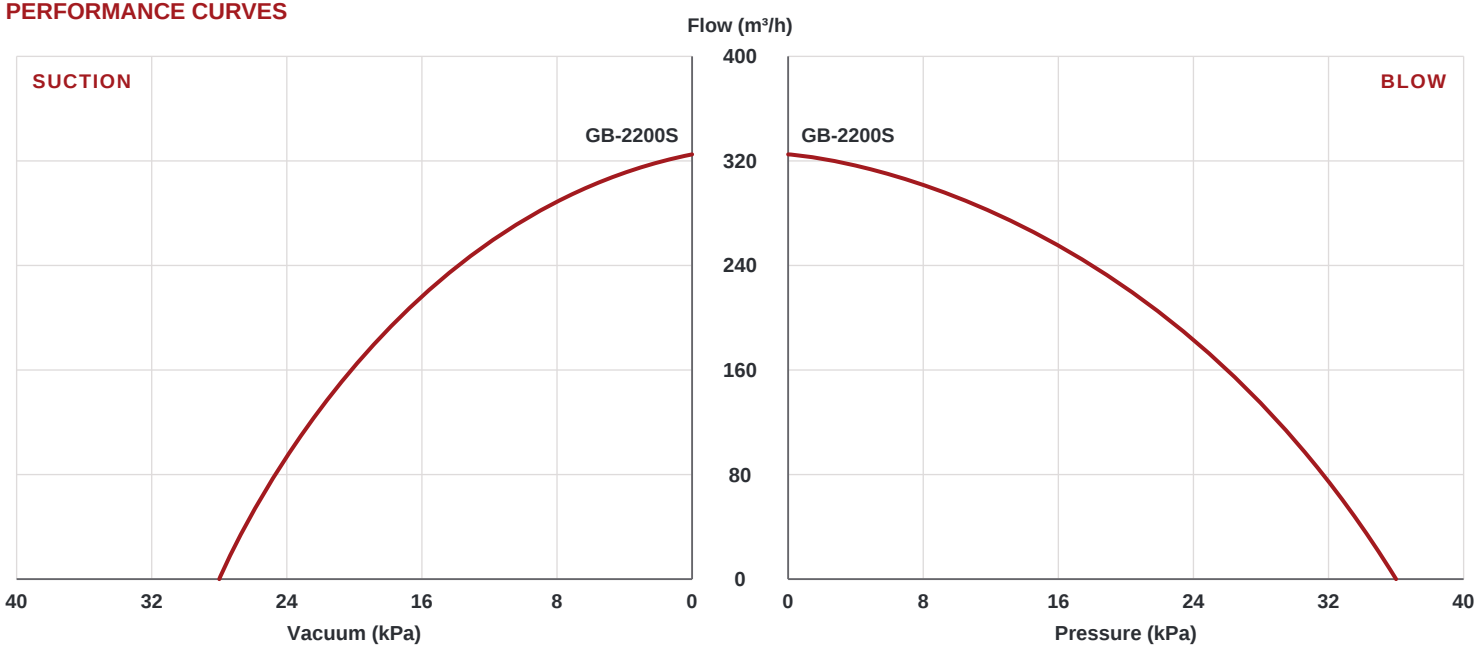
GB-2200
Single-phase 1~ 220–240 V

PERFORMANCE & TECHNICAL DATA

50 Hz | n ≈ 2800 /min

Redpump
Italy Standard

PERFORMANCE CURVES



TECHNICAL DATA

MODEL	POWER kW	POWER HP	RATED CURRENT A	MAX. FLOW m³/h	MAX. PRESSURE kPa	MAX. VACUUM kPa	NOISE dB
THREE-PHASE 3~ 380–415 V							
GB-2200S	2.2	3	4.92	325	36	28	< 70
SINGLE-PHASE 1~ 220–240 V							
GB-2200	2.2	3	14.8	325	36	28	< 70

OPERATING PRINCIPLE

The impellers are mounted directly on the motor shaft for compression without contact or friction. Maximum operational reliability, even at high differential pressure, is ensured by bearings located outside the compression chamber.

NOTES

Performance curves of the GB Series vortex air pumps are for reference only and are not a basis for testing.

Curves GB-2200S also apply to GB-2200 (1~).

Rated current at 220 V (1~) and 380 V (3~).

GB SERIES

GB-3000
GB-4000

High-pressure vortex air pump



RANGE CHARACTERISTICS

Power	**3 – 4 kW**
Max. flow	**up to 475 m³/h**
Max. pressure	**up to 38 kPa**
Max. vacuum	**up to 30 kPa**
Speed	**n ≈ 2800 /min**
Noise	**< 70 dB**

MODEL CODE

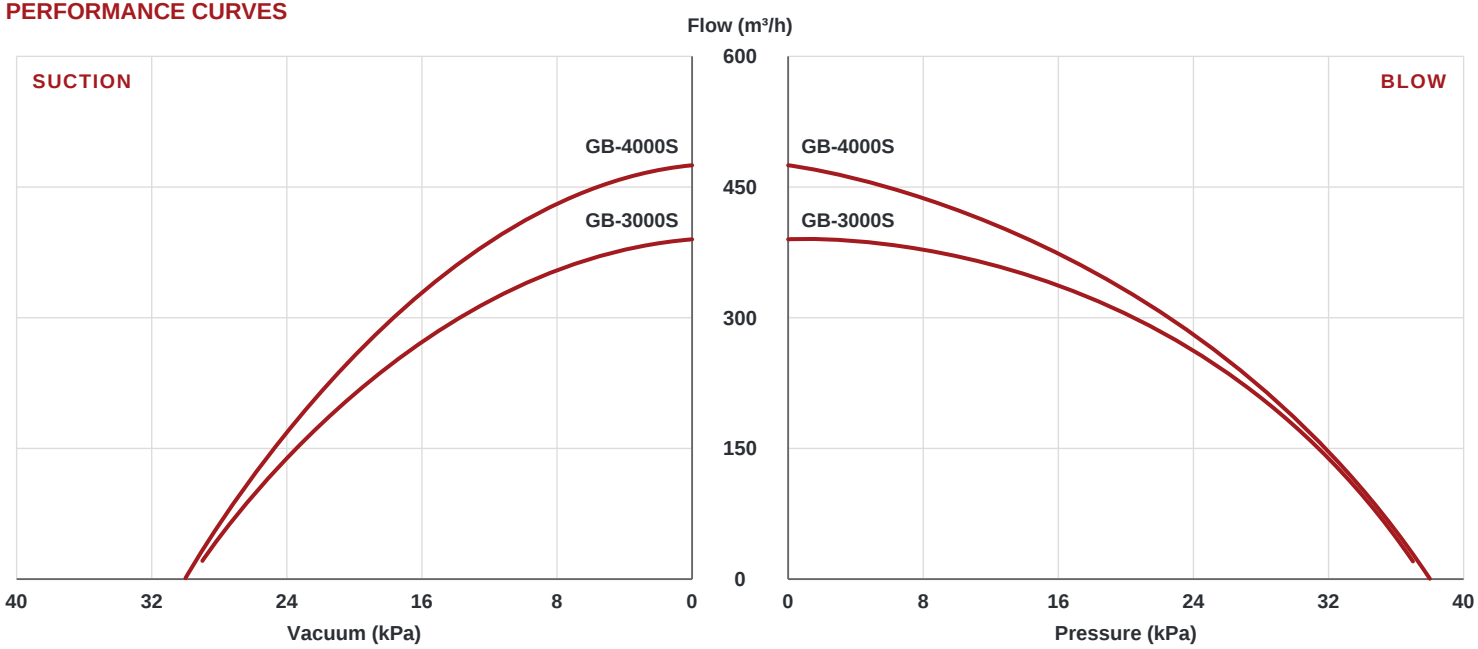
GB-3000S · GB-4000S
Three-phase 3~ 380–415 V

PERFORMANCE & TECHNICAL DATA

50 Hz | n ≈ 2800 /min

Redpump
Italy Standard

PERFORMANCE CURVES



TECHNICAL DATA

MODEL	POWER kW	POWER HP	RATED CURRENT A	MAX. FLOW m³/h	MAX. PRESSURE kPa	MAX. VACUUM kPa	NOISE dB
THREE-PHASE 3~ 380–415 V							
GB-3000S	3	4	6.39	390	37	29	< 70
GB-4000S	4	5.5	8.17	475	38	30	< 70

OPERATING PRINCIPLE

The impellers are mounted directly on the motor shaft for compression without contact or friction. Maximum operational reliability, even at high differential pressure, is ensured by bearings located outside the compression chamber.

NOTES

Performance curves of the GB Series vortex air pumps are for reference only and are not a basis for testing.

Rated current at 380 V (3~).

GB SERIES

GB-5500
GB-7500

High-pressure vortex air pump



RANGE CHARACTERISTICS

Power	**5.5 – 7.5 kW**
Max. flow	**up to 550 m³/h**
Max. pressure	**up to 45 kPa**
Max. vacuum	**up to 35 kPa**
Speed	**n ≈ 2800 /min**
Noise	**< 70 dB**

MODEL CODE

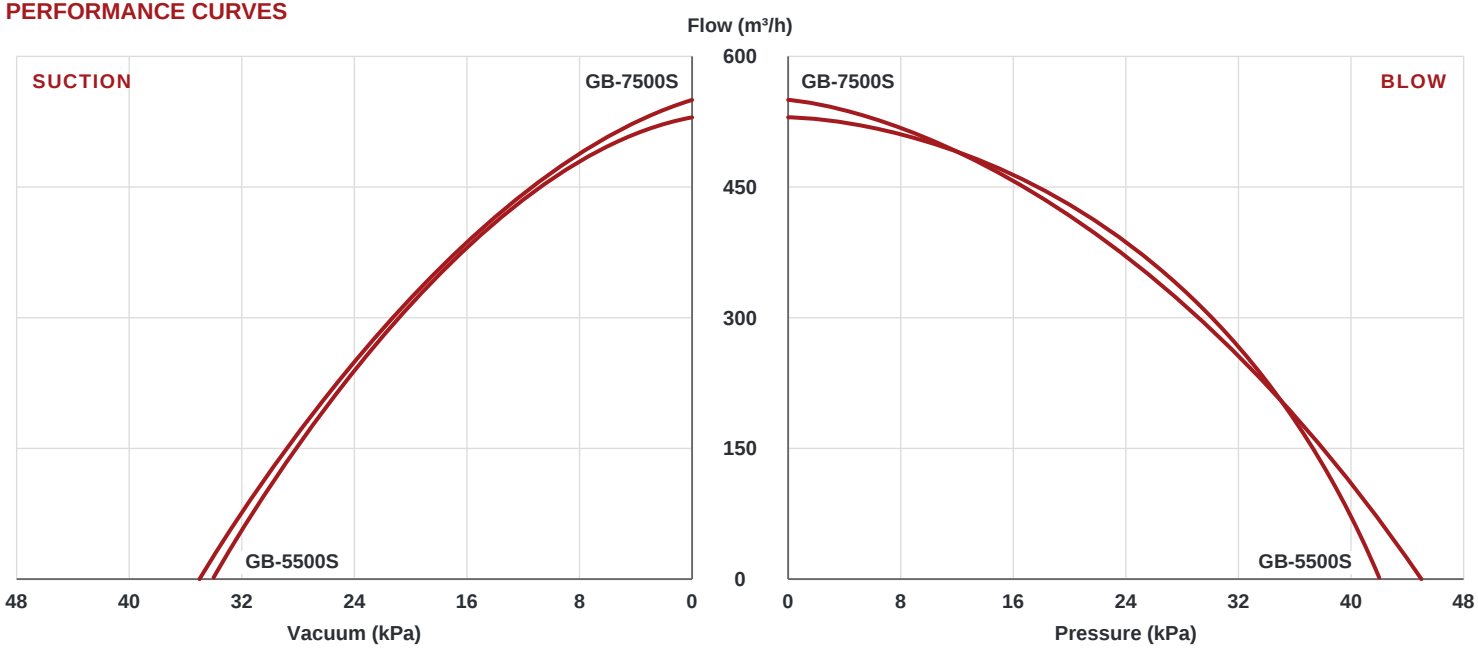
GB-5500S · GB-7500S
Three-phase 3~ 380–415 V

PERFORMANCE & TECHNICAL DATA

50 Hz | n ≈ 2800 /min

Redpump
Italy Standard

PERFORMANCE CURVES



TECHNICAL DATA

MODEL	POWER kW	POWER HP	RATED CURRENT A	MAX. FLOW m³/h	MAX. PRESSURE kPa	MAX. VACUUM kPa	NOISE dB
THREE-PHASE 3~ 380–415 V							
GB-5500S	5.5	7.5	11.1	530	42	34	< 70
GB-7500S	7.5	10	15	550	45	35	< 70

OPERATING PRINCIPLE

The impellers are mounted directly on the motor shaft for compression without contact or friction. Maximum operational reliability, even at high differential pressure, is ensured by bearings located outside the compression chamber.

NOTES

Performance curves of the GB Series vortex air pumps are for reference only and are not a basis for testing.

Rated current at 380 V (3~).

GB SERIES

GB-9500
GB-11000

High-pressure vortex air pump



RANGE CHARACTERISTICS

Power	**9.5 – 11 kW**
Max. flow	**up to 1050 m³/h**
Max. pressure	**up to 75 kPa**
Max. vacuum	**up to 40 kPa**
Speed	**n ≈ 2800 /min**
Noise	**< 70 dB**

MODEL CODE

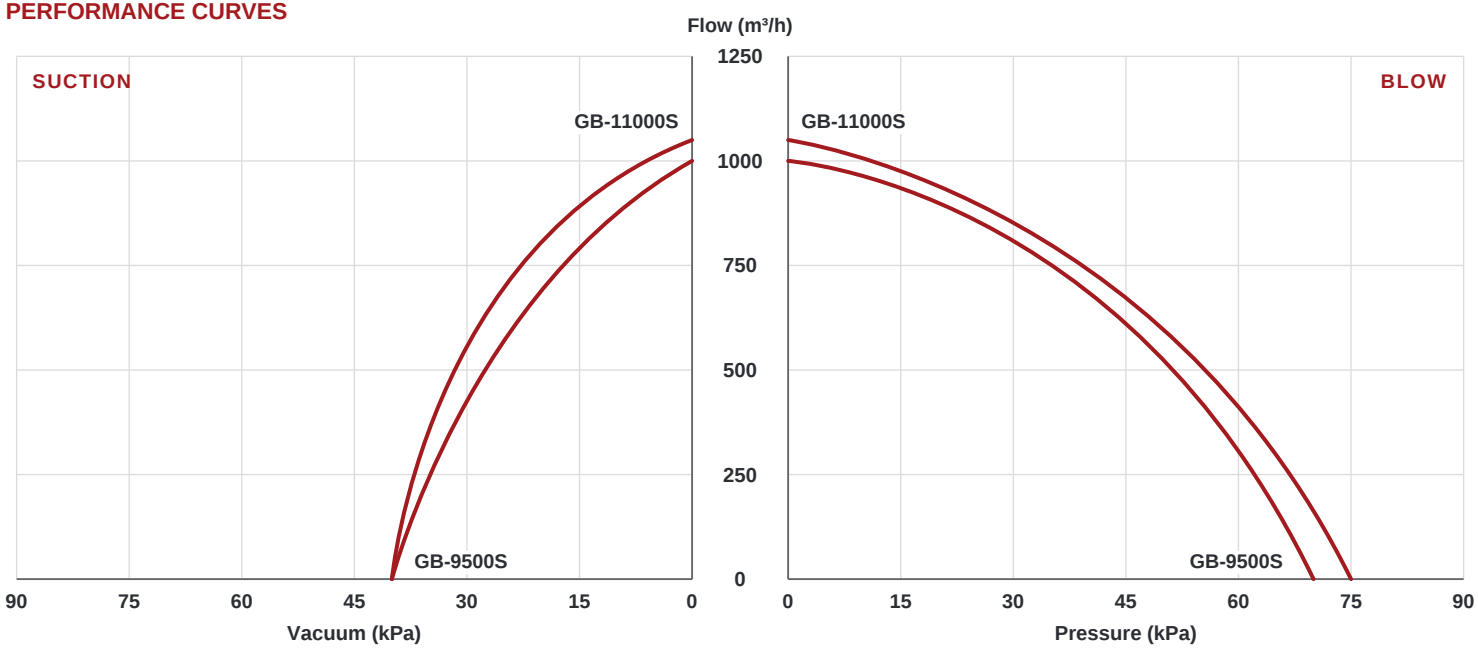
GB-9500S · GB-11000S
Three-phase 3~ 380–415 V

PERFORMANCE & TECHNICAL DATA

50 Hz | n ≈ 2800 /min

Redpump
Italy Standard

PERFORMANCE CURVES



TECHNICAL DATA

MODEL	POWER kW	POWER HP	RATED CURRENT A	MAX. FLOW m³/h	MAX. PRESSURE kPa	MAX. VACUUM kPa	NOISE dB
THREE-PHASE 3~ 380–415 V							
GB-9500S	9.5	12.5	18	1000	70	40	< 70
GB-11000S	11	15	21	1050	75	40	< 70

OPERATING PRINCIPLE

The impellers are mounted directly on the motor shaft for compression without contact or friction. Maximum operational reliability, even at high differential pressure, is ensured by bearings located outside the compression chamber.

NOTES

Performance curves of the GB Series vortex air pumps are for reference only and are not a basis for testing.

Rated current at 380 V (3~).

GB SERIES

GB-15000
GB-18000

High-pressure vortex air pump



RANGE CHARACTERISTICS

Power	**15 – 18 kW**
Max. flow	**up to 1050 m³/h**
Max. pressure	**up to 79 kPa**
Max. vacuum	**up to 40 kPa**
Speed	**n ≈ 2800 /min**
Noise	**< 70 dB**

MODEL CODE

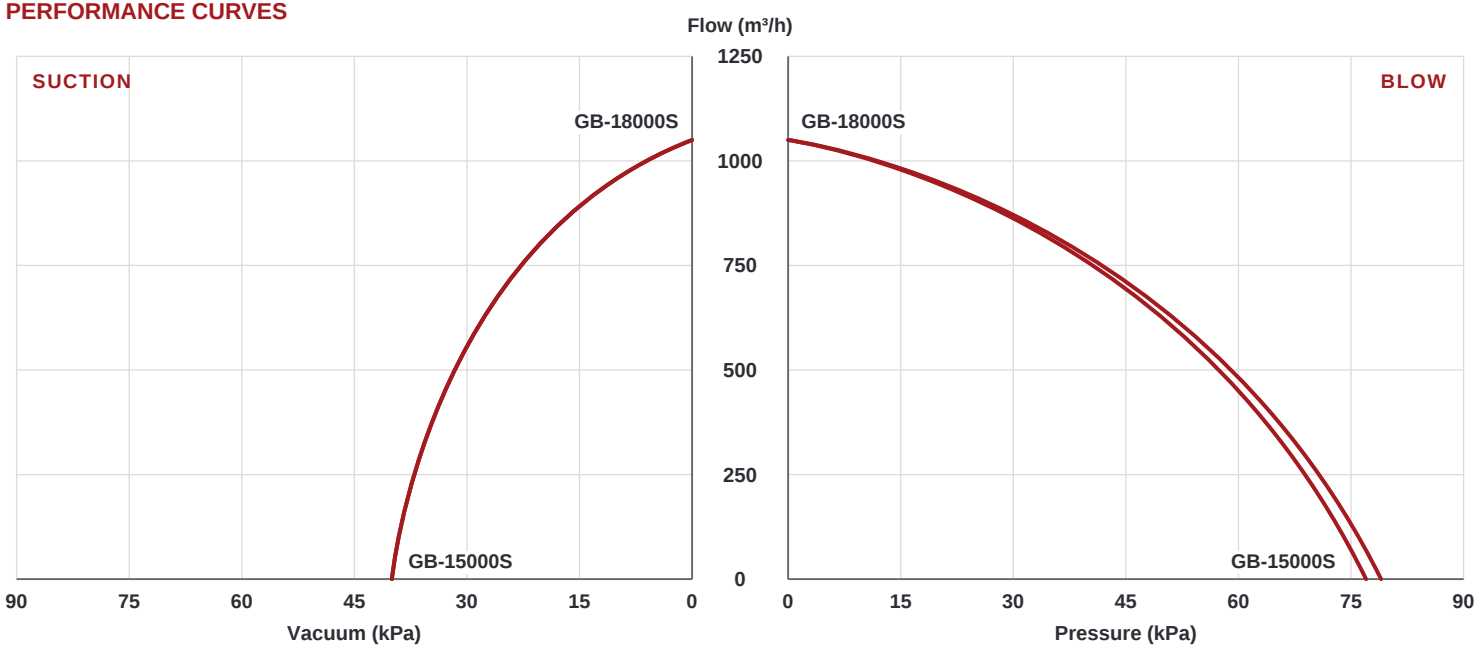
GB-15000S · GB-18000S
Three-phase 3~ 380–415 V

PERFORMANCE & TECHNICAL DATA

50 Hz | n ≈ 2800 /min

Redpump
Italy Standard

PERFORMANCE CURVES



TECHNICAL DATA

MODEL	POWER kW	POWER HP	RATED CURRENT A	MAX. FLOW m³/h	MAX. PRESSURE kPa	MAX. VACUUM kPa	NOISE dB
THREE-PHASE 3~ 380–415 V							
GB-15000S	15	20	28	1050	77	40	< 70
GB-18000S	18	24	34	1050	79	40	< 70

OPERATING PRINCIPLE

The impellers are mounted directly on the motor shaft for compression without contact or friction. Maximum operational reliability, even at high differential pressure, is ensured by bearings located outside the compression chamber.

NOTES

Performance curves of the GB Series vortex air pumps are for reference only and are not a basis for testing.

Rated current at 380 V (3~).

GB SERIES

GB-550/2
GB-750/2

Double-stage vortex air pump



RANGE CHARACTERISTICS

Power	**0.55 – 0.75 kW**
Max. flow	**up to 88 m³/h**
Max. pressure	**up to 31 kPa**
Max. vacuum	**up to 25 kPa**
Noise	**< 60 dB**

MODEL CODE

GB-550S/2 · GB-750S/2
Three-phase 3~ 380–415 V

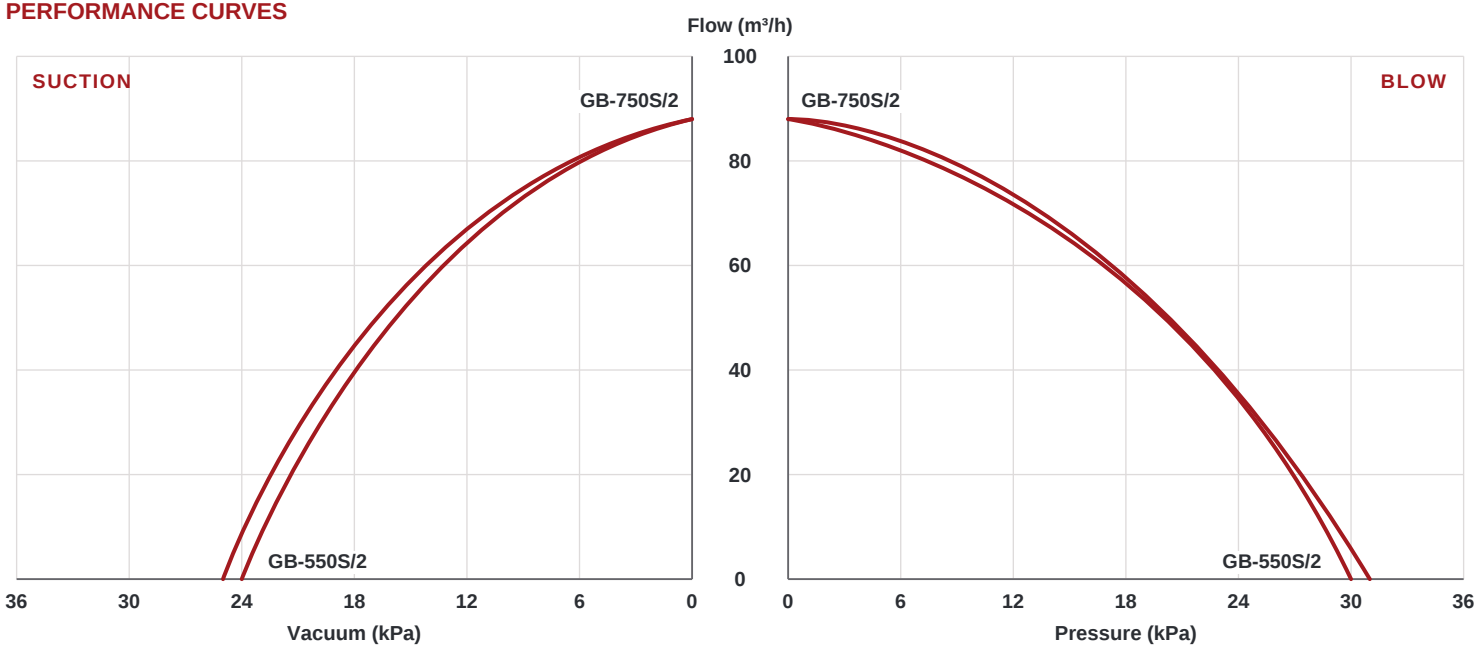
GB-550/2 · GB-750/2
Single-phase 1~ 220–240 V

PERFORMANCE & TECHNICAL DATA

50 Hz

Redpump
Italy Standard

PERFORMANCE CURVES



TECHNICAL DATA

MODEL	POWER kW	POWER HP	RATED CURRENT A	MAX. FLOW m³/h	MAX. PRESSURE kPa	MAX. VACUUM kPa	NOISE dB
THREE-PHASE 3~ 380–415 V							
GB-550S/2	0.55	0.75	2.2	88	30	24	< 60
GB-750S/2	0.75	1	2.4	88	31	25	< 60
SINGLE-PHASE 1~ 220–240 V							
GB-550/2	0.55	0.75	4.8	88	30	24	< 60
GB-750/2	0.75	1	6.1	88	31	25	< 60

OPERATING PRINCIPLE

The impellers are mounted directly on the motor shaft for compression without contact or friction. Maximum operational reliability, even at high differential pressure, is ensured by bearings located outside the compression chamber.

NOTES

Performance curves of the GB Series vortex air pumps are for reference only and are not a basis for testing.

Curves GB-550S/2 / GB-750S/2 also apply to GB-550/2 / GB-750/2 (1~).

Rated current at 220 V (1~) and 380 V (3~).

GB SERIES

GB-1100/2
GB-1500/2

Double-stage vortex air pump



RANGE CHARACTERISTICS

Power	**1.1 – 1.5 kW**
Max. flow	**up to 145 m³/h**
Max. pressure	**up to 38 kPa**
Max. vacuum	**up to 31 kPa**
Noise	**< 60 dB**

MODEL CODE

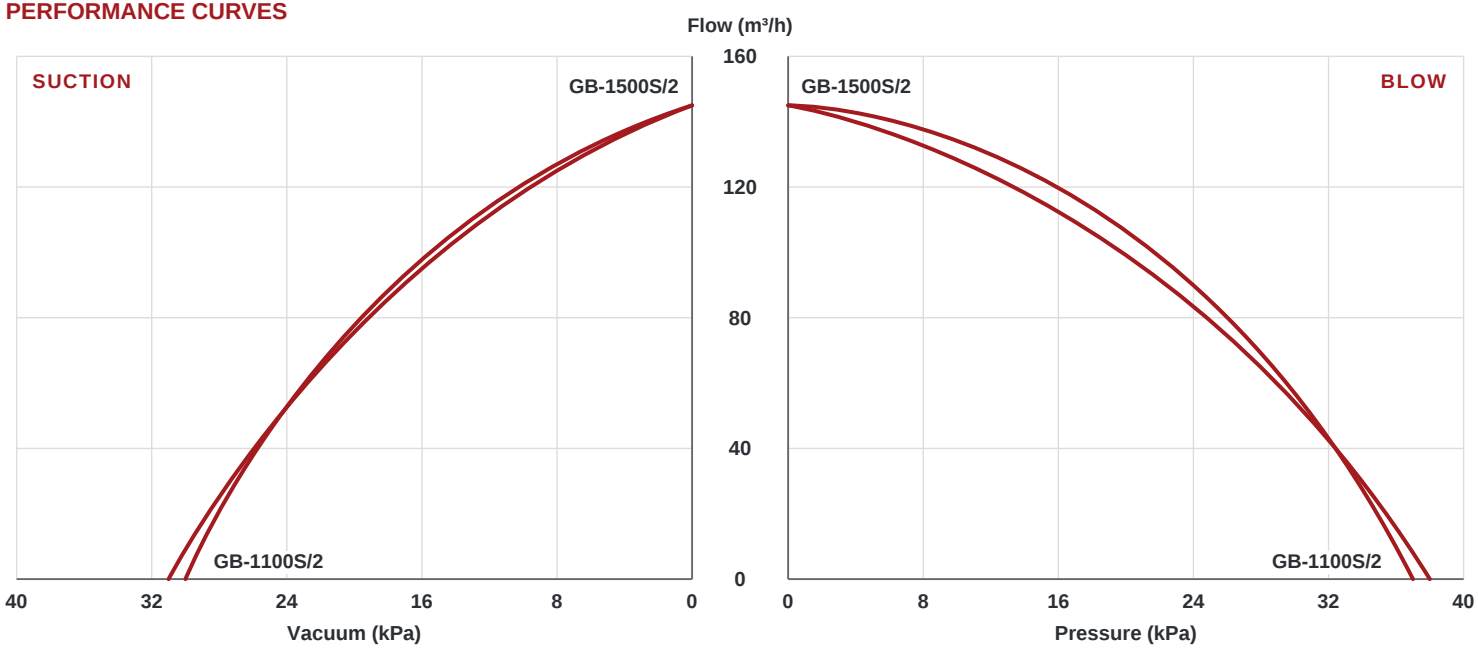
GB-1100S/2 · GB-1500S/2
Three-phase 3~ 380–415 V

GB-1100/2 · GB-1500/2
Single-phase 1~ 220–240 V

PERFORMANCE & TECHNICAL DATA

50 Hz

PERFORMANCE CURVES



TECHNICAL DATA

MODEL	POWER kW	POWER HP	RATED CURRENT A	MAX. FLOW m³/h	MAX. PRESSURE kPa	MAX. VACUUM kPa	NOISE dB
THREE-PHASE 3~ 380–415 V							
GB-1100S/2	1.1	1.5	3.3	145	37	30	< 60
GB-1500S/2	1.5	2	4.3	145	38	31	< 60
SINGLE-PHASE 1~ 220–240 V							
GB-1100/2	1.1	1.5	8.4	145	37	30	< 60
GB-1500/2	1.5	2	11	145	38	31	< 60

OPERATING PRINCIPLE

The impellers are mounted directly on the motor shaft for compression without contact or friction. Maximum operational reliability, even at high differential pressure, is ensured by bearings located outside the compression chamber.

NOTES

Performance curves of the GB Series vortex air pumps are for reference only and are not a basis for testing.

Curves GB-1100S/2 / GB-1500S/2 also apply to GB-1100/2 / GB-1500/2 (1~).

Rated current at 220 V (1~) and 380 V (3~).

GB SERIES

GB-2200/2
GB-3000/2

Double-stage vortex air pump



RANGE CHARACTERISTICS

Power	**2.2 – 3 kW**
Max. flow	**up to 220 m³/h**
Max. pressure	**up to 44 kPa**
Max. vacuum	**up to 43 kPa**
Noise	**< 70 dB**

MODEL CODE

GB-2200S/2 · GB-3000S/2
Three-phase 3~ 380–415 V

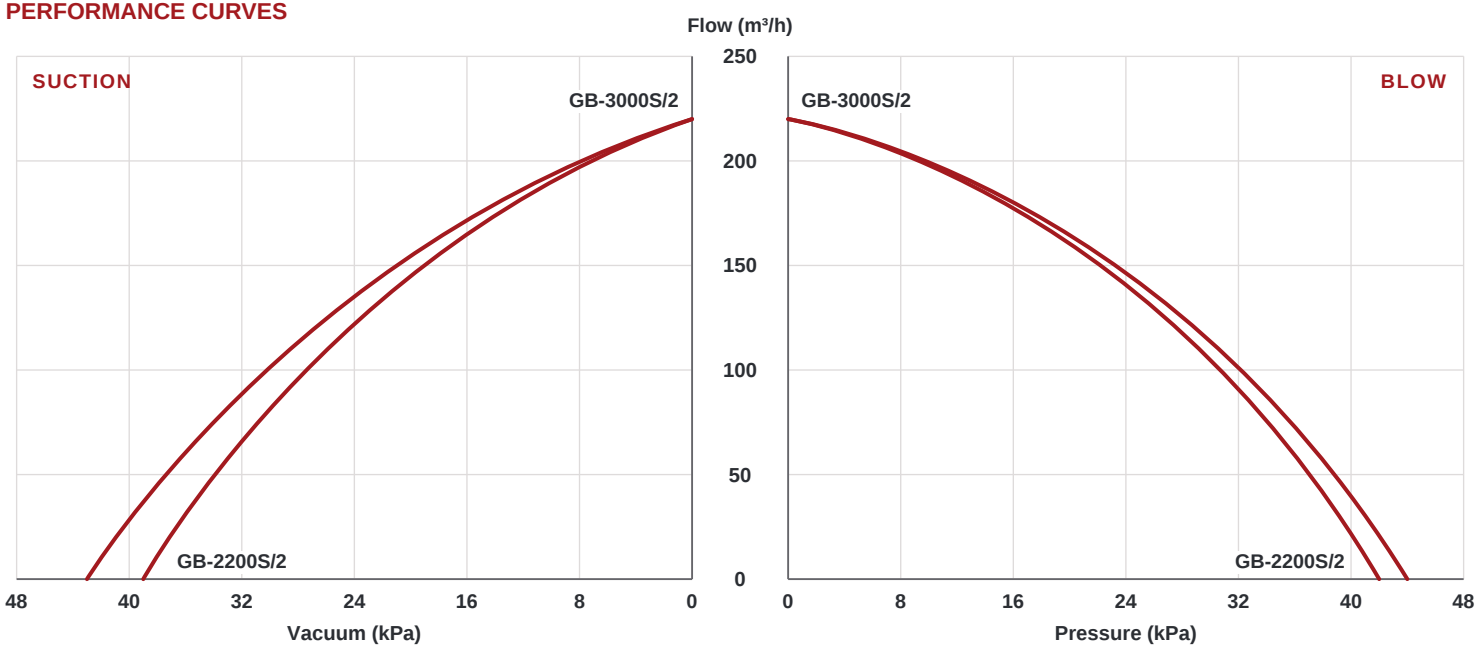
GB-2200/2
Single-phase 1~ 220–240 V

PERFORMANCE & TECHNICAL DATA

50 Hz

Redpump
Italy Standard

PERFORMANCE CURVES



TECHNICAL DATA

MODEL	POWER kW	POWER HP	RATED CURRENT A	MAX. FLOW m³/h	MAX. PRESSURE kPa	MAX. VACUUM kPa	NOISE dB
THREE-PHASE 3~ 380–415 V							
GB-2200S/2	2.2	3	5.6	220	42	39	< 70
GB-3000S/2	3	4	7.2	220	44	43	< 70
SINGLE-PHASE 1~ 220–240 V							
GB-2200/2	2.2	3	15.5	220	42	39	< 70

OPERATING PRINCIPLE

The impellers are mounted directly on the motor shaft for compression without contact or friction. Maximum operational reliability, even at high differential pressure, is ensured by bearings located outside the compression chamber.

NOTES

Performance curves of the GB Series vortex air pumps are for reference only and are not a basis for testing.

Curves GB-2200S/2 also apply to GB-2200/2 (1~).

Rated current at 220 V (1~) and 380 V (3~).

GB SERIES

GB-4000/2
GB-5500/2

Double-stage vortex air pump



RANGE CHARACTERISTICS

Power	**4 – 5.5 kW**
Max. flow	**up to 325 m³/h**
Max. pressure	**up to 63 kPa**
Max. vacuum	**up to 48 kPa**
Noise	**< 70 dB**

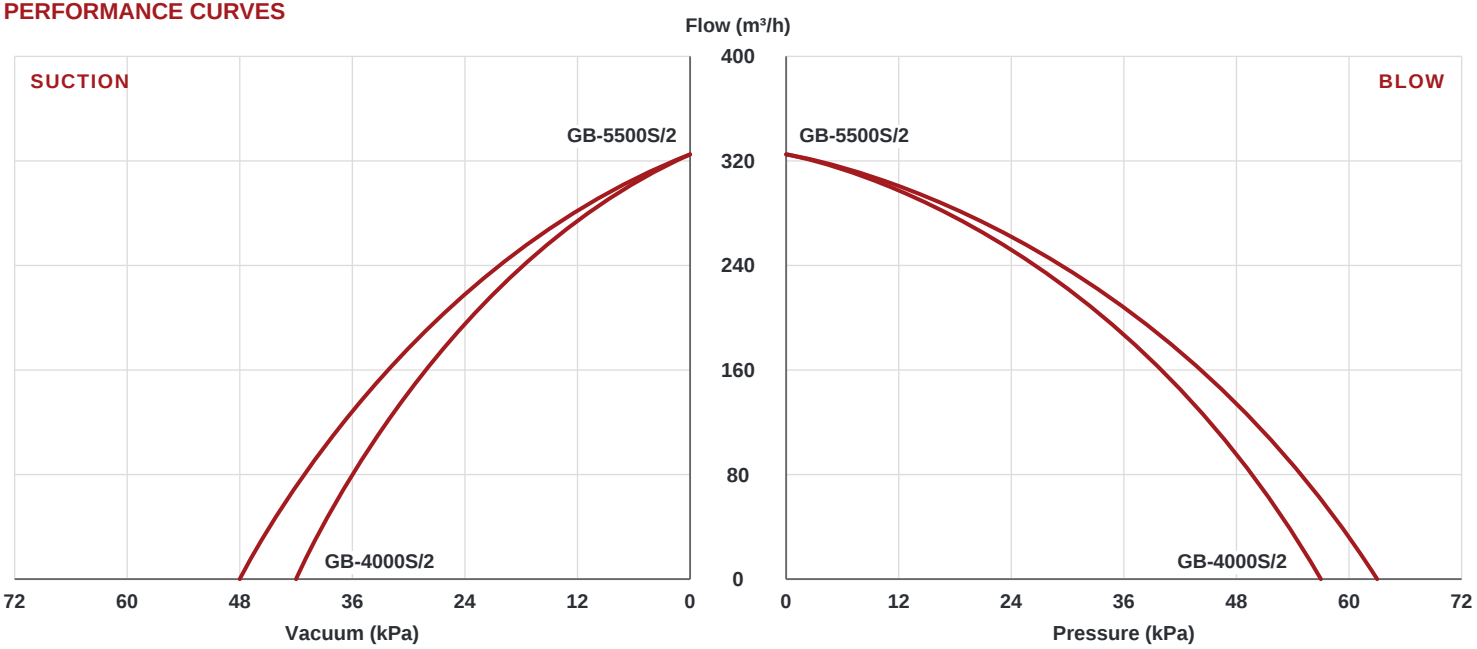
MODEL CODE

GB-4000S/2 · GB-5500S/2
Three-phase 3~ 380–415 V

PERFORMANCE & TECHNICAL DATA

50 Hz

PERFORMANCE CURVES



TECHNICAL DATA

MODEL	POWER kW	POWER HP	RATED CURRENT A	MAX. FLOW m³/h	MAX. PRESSURE kPa	MAX. VACUUM kPa	NOISE dB
THREE-PHASE 3~ 380–415 V							
GB-4000S/2	4	5.5	9	325	57	42	< 70
GB-5500S/2	5.5	7.5	13.3	325	63	48	< 70

OPERATING PRINCIPLE

The impellers are mounted directly on the motor shaft for compression without contact or friction. Maximum operational reliability, even at high differential pressure, is ensured by bearings located outside the compression chamber.

NOTES

Performance curves of the GB Series vortex air pumps are for reference only and are not a basis for testing.

Rated current at 380 V (3~).

GB SERIES

GB-7500/2
GB-9500/2

Double-stage vortex air pump



RANGE CHARACTERISTICS

Power	**7.5 – 9.5 kW**
Max. flow	**up to 520 m³/h**
Max. pressure	**up to 70 kPa**
Max. vacuum	**up to 55 kPa**
Noise	**< 80 dB**

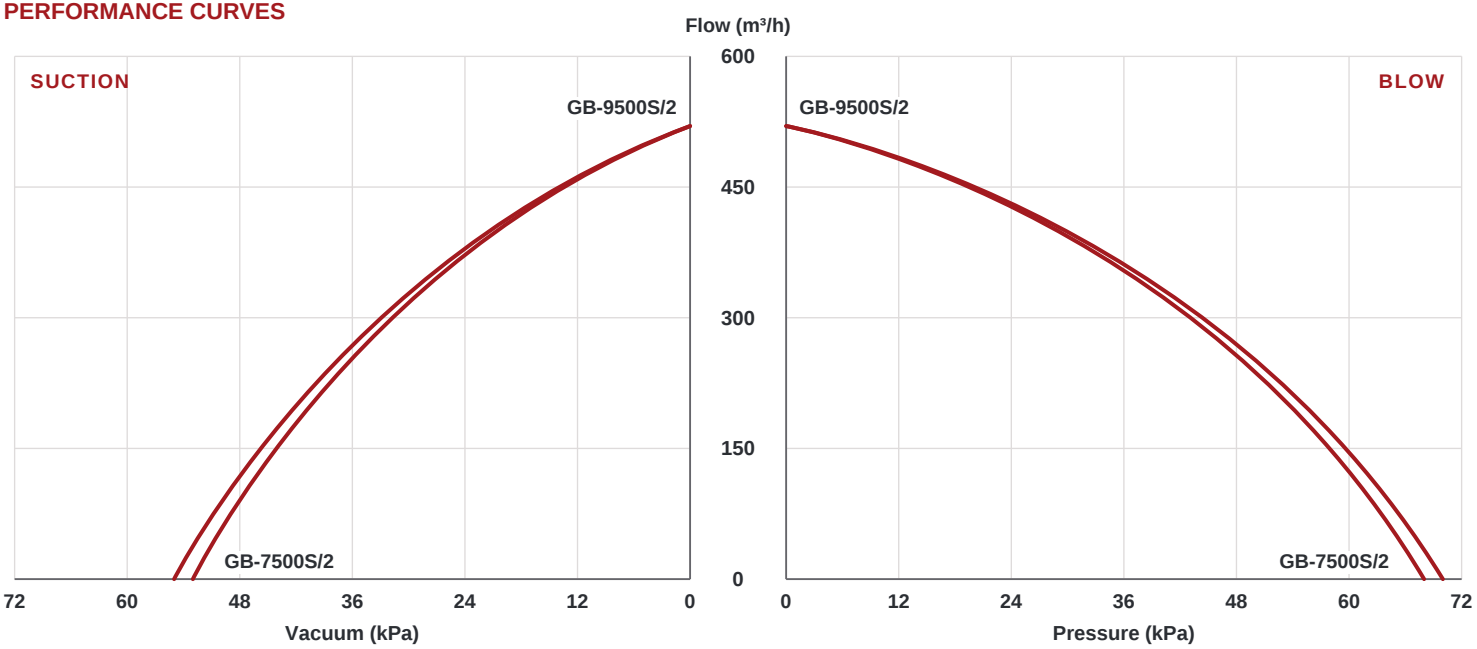
MODEL CODE

GB-7500S/2 · GB-9500S/2
Three-phase 3~ 380–415 V

PERFORMANCE & TECHNICAL DATA

50 Hz

PERFORMANCE CURVES



TECHNICAL DATA

MODEL	POWER kW	POWER HP	RATED CURRENT A	MAX. FLOW m³/h	MAX. PRESSURE kPa	MAX. VACUUM kPa	NOISE dB
THREE-PHASE 3~ 380–415 V							
GB-7500S/2	7.5	10	16.7	520	68	53	< 80
GB-9500S/2	9.5	12.5	17.3	520	70	55	< 80

OPERATING PRINCIPLE

The impellers are mounted directly on the motor shaft for compression without contact or friction. Maximum operational reliability, even at high differential pressure, is ensured by bearings located outside the compression chamber.

NOTES

Performance curves of the GB Series vortex air pumps are for reference only and are not a basis for testing.

Rated current at 380 V (3~).

GB SERIES

GB-11000/2
GB-15000/2

Double-stage vortex air pump



RANGE CHARACTERISTICS

Power	**11 – 15 kW**
Max. flow	**up to 560 m³/h**
Max. pressure	**up to 72 kPa**
Max. vacuum	**up to 57 kPa**
Noise	**< 80 dB**

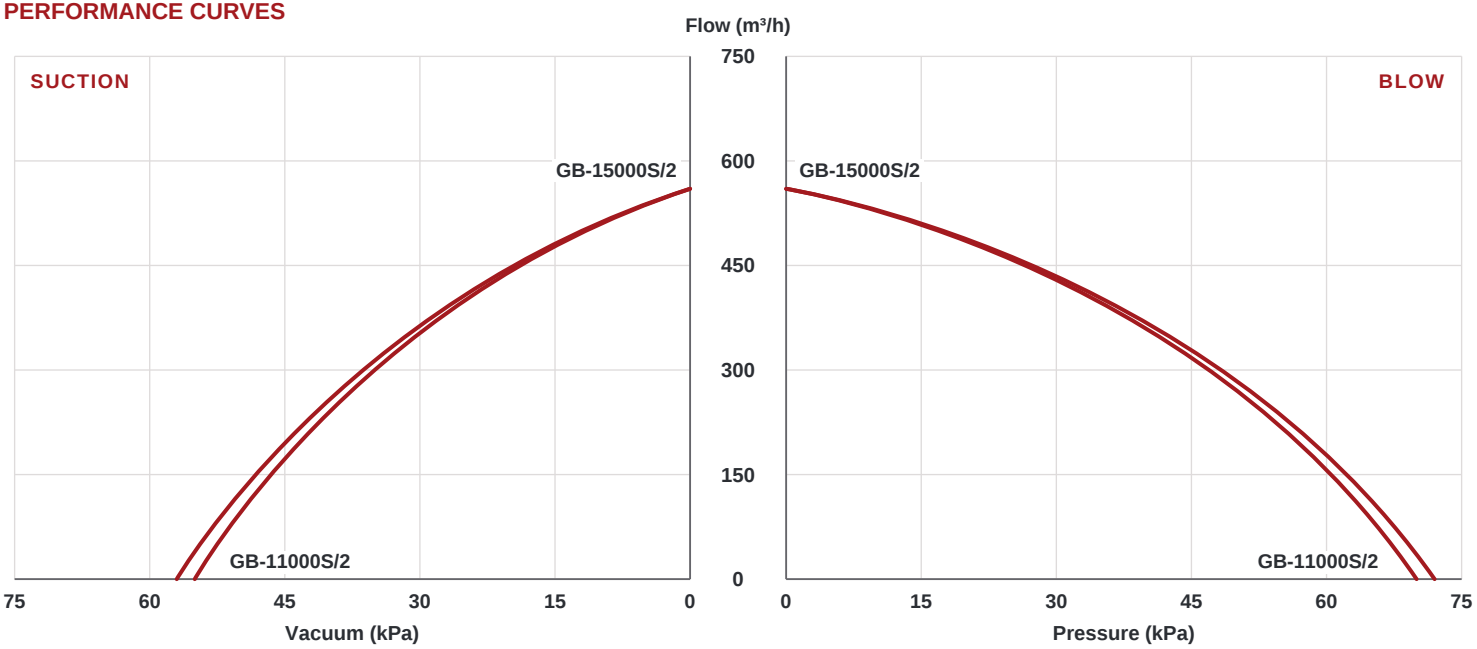
MODEL CODE

GB-11000S/2 · GB-15000S/2
Three-phase 3~ 380–415 V

PERFORMANCE & TECHNICAL DATA

50 Hz

PERFORMANCE CURVES



TECHNICAL DATA

MODEL	POWER kW	POWER HP	RATED CURRENT A	MAX. FLOW m³/h	MAX. PRESSURE kPa	MAX. VACUUM kPa	NOISE dB
THREE-PHASE 3~ 380–415 V							
GB-11000S/2	11	15	21.5	560	70	55	< 80
GB-15000S/2	15	20	32.5	560	72	57	< 80

OPERATING PRINCIPLE

The impellers are mounted directly on the motor shaft for compression without contact or friction. Maximum operational reliability, even at high differential pressure, is ensured by bearings located outside the compression chamber.

NOTES

Performance curves of the GB Series vortex air pumps are for reference only and are not a basis for testing.

Rated current at 380 V (3~).

GB SERIES

FITTINGS

Vortex air pump fittings

RANGE

01 CARTRIDGE FILTER
02 MESH FILTER CAP
03 RELIEF VALVE
04 DISTRIBUTOR
05 SILENCER
06 FILTER HOUSING
07 AIR KNIFE
08 FILTER + PIPE
09 PIPE
10 ELBOW PIPE

VORTEX AIR PUMP FITTINGS

GB Series



01 CARTRIDGE FILTER



02 MESH FILTER CAP



03 RELIEF VALVE



04 DISTRIBUTOR



05 SILENCER



06 FILTER HOUSING



07 AIR KNIFE



08 FILTER + PIPE



09 PIPE



10 ELBOW PIPE